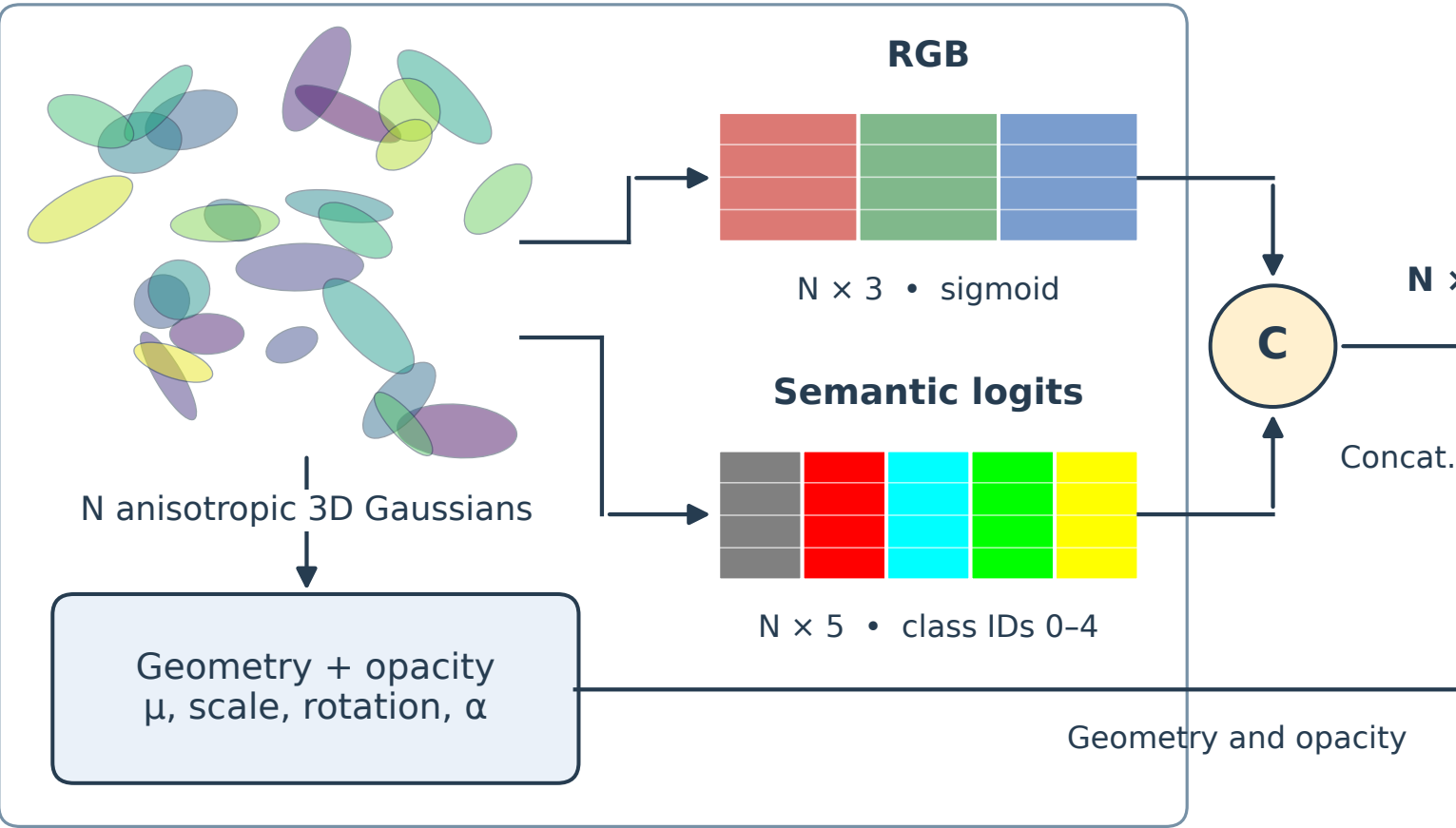


Learnable Gaussians Per-Gaussian attributes



Shared renderer

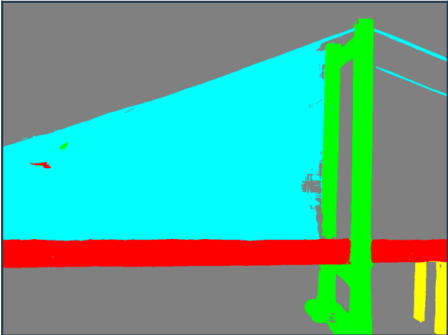
Camera: K, R, t

Differentiable rasterization
Shared α -compositing

RGB image
 $H \times W \times 3$



Semantic map



Rendered logits
 $H \times W \times 5$

Argmax

Labeled / pseudo masks

RGB targets

Semantic loss
Cross-entropy

Photometric loss
 $0.8 L_1 + 0.2 (1 - \text{SSIM})$

All Gaussian parameters

Photometric: geometry, opacity, RGB
Semantic: geometry, opacity, logits

Backpropagation

$$\mathcal{L} = \mathcal{L}_{\text{photo}} + \lambda_{\text{sem}} w \mathcal{L}_{\text{CE}}$$

w: mask-source weight

